

SAMEER RAI

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SUMMARY

Computer Engineering student with strong foundations in **Data Structures & Algorithms** and **backend development**. Proficient in **Java** and **Python**, with experience building **scalable systems** and **REST APIs**. Interested in **system design**, **problem solving**, and **writing efficient, reliable code**.

EDUCATION

Thapar Institute of Engineering and Technology

BTech in Computer Engineering. CGPA : 8.68

St. Joseph Montessori School

ISC : 96.00% ICSE : 95.60%

Patiala, Punjab

Aug 2023 – June 2027

Lucknow, Uttar Pradesh

Mar 2015 – Feb 2022

EXPERIENCE

Undergraduate Research Assistant, Thapar University

Jun 2025 – Sep 2025

- Analyzed **MRI-based brain tumor detection** using BRATS dataset (**1K+ scans**), developing preprocessing pipelines for medical image analysis
- Benchmarked **5+ CNN architectures** for tumor segmentation and prognosis prediction, comparing performance across models

PROJECTS

API Monitoring Dashboard 🔗

Technologies: *Java, Spring Boot, React.js, REST APIs*

Feb 2026

- Engineered a **full-stack API monitoring platform** tracking **latency, uptime, and HTTP status** across **5+ endpoints** in realtime.
- Designed **health-check mechanisms** and built **interactive dashboards** to enable faster failure detection and visualize system performance.

Disaster Intelligence System 🔗

Technologies: *Python, FastAPI, Scikit-learn, Gemini API, REST APIs, React.js*

Dec 2025

- Designed a **multimodal ML pipeline** combining **BiLSTM (GloVe)** for text filtering and **ResNet-50** for image classification, achieving **93% image accuracy** and **82% text accuracy**.
- Processed **10K+ tweets** and **13K+ images** with a **two-stage conditional pipeline**, reducing unnecessary computation and enabling **real-time disaster detection and alerting**.
- Integrated **FastAPI inference**, automated **report generation**, and **email-based alerting** for end-to-end deployment.

TOPSIS Decision Analysis Package (PyPI) 🔗

Technologies: *Python, NumPy, Pandas, PyPI, Flask*

Jan 2026

- Published **Topsis-Sameer-102303773** on PyPI, a pip-installable Python package implementing the **TOPSIS algorithm** for multi-criteria decision making.
- Built **CLI utilities** and a **Flask-based web interface** to automate ranking of alternatives from CSV datasets.

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Database Management System
- Artificial Intelligence
- Operating System
- Object-Oriented Programming
- Data Science : Predictive Analytics

TECHNICAL SKILLS

Languages: Python, Java, C/C++, R, JavaScript, SQL

Frameworks: Spring Boot, React.js, FastAPI, Flask, Scikit-learn

Databases: MySQL, MongoDB

Developer Tools: Git, Docker, Linux (Ubuntu), VS Code, PyCharm, Google Colab

ACHIEVEMENTS

- Code Marathon 🔗** — Secured **1st Runner-Up** among 80+ teams in a relay-style coding competition

CERTIFICATES AND SHORT COURSES

- Unit Testing in Java 🔗** : JUnit, Service Testing.
- CI/CD Pipeline with Docker: From Code to Deployment 🔗** : Docker, CI/CD Pipelines.
- Open Source Model with Hugging Face 🔗** : MultiModal, Natural Language Processing (NLP), Transformers.
- Full Stack Development Course 🔗** : MERN Development.